

Q1. Check if a Number is Power of 2

Problem:

Write a function to check whether a given number is a power of 2 using bit manipulation.

Example:

Input: 16

Output: true

Input: 18

Output: false

Focus:

- $n \& (n - 1)$ trick
 - Binary representation thinking
 - $O(1)$ solution
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Q2. Count Set Bits

Problem:

Write a function to count the number of 1s in the binary representation of a number.

Example:

Input: 13

Output: 3

(13 $\rightarrow$ 1101)

Focus:

- Right shift operator
 - $n \& 1$ check
 - Brian Kernighan's Algorithm
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Q3. Find Unique Element

Problem:

In an array where every element appears twice except one, find the unique element.

Example:

Input: [4, 1, 2, 1, 2]

Output: 4

Focus:

- XOR properties
- $a \wedge a = 0$
- $O(n)$ time, $O(1)$ space

Q4. Generate All Subsets using Bit Masking

Problem:

Given an array of n elements, generate all possible subsets using bit masking.

Example:

Input: [1, 2]

Output:

[]

[1]

[2]

[1,2]

Focus:

- 2^n subsets
 - Bit representation of numbers from 0 to $(2^n - 1)$
 - Mask checking
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Q5. Merge Two Sorted Arrays

Problem:

Write a function to merge two sorted arrays into one sorted array.

Example:

Input:

[1, 3, 5]

[2, 4, 6]

Output:

[1, 2, 3, 4, 5, 6]

Focus:

- Two pointer technique
 - Stability of Merge Sort
 - $O(n)$ merging logic
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Q6. Implement Merge Sort

Problem:

Implement Merge Sort to sort an array.

Example:

Input: [5, 2, 4, 1]

Output: [1, 2, 4, 5]

Focus:

- Divide array into halves
- Recursion
- Time Complexity: $O(n \log n)$
- Space Complexity: $O(n)$

Q7. Implement Quick Sort

Problem:

Implement Quick Sort to sort an array using partition logic.

Example:

Input: [10, 7, 8, 9, 1, 5]

Output: [1, 5, 7, 8, 9, 10]

Focus:

- Pivot selection
 - Partition algorithm
 - Average: $O(n \log n)$
 - Worst Case: $O(n^2)$
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Q8. Kth Smallest Element using Quick Sort Partition

Problem:

Find the kth smallest element in an array using Quick Select (partition logic).

Example:

Input: [7, 10, 4, 3, 20, 15], $k = 3$

Output: 7

Focus:

- Partition logic reuse
 - Quick Select
 - Average $O(n)$
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Q9. Count Inversions in an Array

Problem:

Count the number of inversions in an array.

Example:

Input: [2, 4, 1, 3, 5]

Output: 3

Focus:

- Modified merge step
- Counting during merge
- $O(n \log n)$ solution

Q10. Check if Two Numbers Differ by Exactly One Bit

Problem:

Write a function to check if two numbers differ by exactly one bit.

Example:

Input: a = 5, b = 4

Binary: 101, 100

Output: true

Focus:

- XOR usage
- Count set bits in (a ^ b)
- Bit difference logic